

Program overview

12-Sep-2023 12:27

Year 2023/2024
Organization Industrial Design Engineering
Education Master Strategic Product Design

Code	Omschrijving	ECTS	p1	p2	p3	p4	p5
Master Strategic Product Design							
1st year Master Strategic Product Design							
General Courses Master Strategic Product Design							
ID4010	Design Theory and Methodology	3					
ID4060	Manage your Master	2					
ID4070	IDE Academy	4					
Master Specific Courses Strategic Product Design							
ID4216	Context and Conceptualisation	6					
ID4315-16	Brand & Product Commercialisation	8					
ID4330	New Product Economics	3					
ID4337	Design Roadmapping	3					
ID4340	Strategic Value of Design	3					
ID4350-16	Design Strategy Project	12					
ID4355-18	SPD Research	13					
ID4370	SPD Media	3					
2nd year Master Strategic Product Design							
ID4390-16	Graduation Project (SPD)	30					
Electives (30 EC) SPD							
Specialization Medisign							
Compulsory Electives (min. 9 ECTS) Medisign							
Compulsory Electives (min. 9 EC), at least 2 EC IDE Electives							
ID5301	Biomechanics	3					
ID5383	Inclusive Design	3					
ID5663	Contextmapping Skills	6					
ID5721	Capita Selecta Medisign	3					
ID5732	Health Psychology: Tools and Methods	3					
ID5741-23	eHealth Design	3					
ID5752	Rules and Regulations for Designing Medical Devices	3					

Year	2023/2024
Organization	Industrial Design Engineering
Education	Master Strategic Product Design

Master Strategic Product Design

Introduction 4

A mandatory application for all courses (mandatory and electives courses) shall be made via the electronic applications system Osiris in the period that Osiris there to is opened. This period closes about 5 calendar weeks prior to the first day of the semester in which the programme starts; the faculty announces the exact deadline for application timely. The application for courses has to take place per semester, meaning for courses of 2 quarters at once.

See <http://www.io.tudelft.nl/osiris> for deadlines and details.

Year	2023/2024
Organization	Industrial Design Engineering
Education	Master Strategic Product Design

1st year Master Strategic Product Design

Year	2023/2024
Organization	Industrial Design Engineering
Education	Master Strategic Product Design

General Courses Master Strategic Product Design

ID4010	Design Theory and Methodology	3
Course Coordinator	Dr. R.S.K. Chandrasegaran	
Contact Hours / Week x/x/x/x	3/0/0/0	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Course Contents	The aim of this course is to help you gain a deeper understanding of the principles, practices and procedures of designing in a broad and general sense. Unlike other Universities, Delft has a rich research and education tradition in design theory and methodology. You have almost certainly heard of design thinking, for example, but did you know that Delft played a key role in its development? Through the course we will cover many topics and approaches but our core principle is to link different types of theory to design practice in a way that is insightful, useful, and entertaining. The course is NOT about learning new methods of designing, it is about reflecting on the concepts, practices, activities, and methods that designing is constituted by. Design Theory and Methodology (DTM) will help you to build a more nuanced vocabulary when talking about the design process thus aiding your development as a designer. Through a number of lectures, critical discussions, and practical activities you will improve your ability to describe, analyse, and understand the activity of designing. This will help you to reflect further on your own design processes, and those of others. Developing an ability to critically analyse design processes will empower you to change, modify, and adapt your thinking and acting while designing.	
Study Goals	By the end of this course it is expected that you will be able to: Learning Objective 1: Articulate theoretical aspects of design practice Learning Objective 2: Critically question design theories, methodologies, and practices Learning Objective 3: Demonstrate a theoretical understanding of your own design practice	
Education Method	The course is centred around a series of lectures and practicums that explore aspects of designing and the design process. A number of activities will help deepen your understanding and coaches will provide support for you at appropriate points. Online resources will help guide you through the course and include: MS Teams discussion forums, podcasts, key references and other relevant materials.	
Literature and Study Materials	The course will be managed on Brightspace with timetable, assignments, and course materials - including podcasts - available. Microsoft Teams will be used to manage discussion and share information during the course.	
Assessment	Assignment 1 The first two weeks of the course are about sensitising you to the concepts of design theory and methodology. Assignment 1 is a multiple-choice quiz to test your comprehension. Assignment 2 Following Assignment 1 you will participate in three practical sessions, which will involve small group work. A reflection on what you did in these sessions, and what you learned, make up your submission for the second assignment. Assignment 3 The final assignment is an individual 1500 word essay and short reflection. The essay is your chance to explore one specific topic area from the course. You will choose an essay topic from a number of options and develop an argument around a theme drawing on references from design research journals. Scholarly work forms the core of the assignment.	
Enrolment / Application	For IDE MSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	
Special Information	Course Coordinators 2022-2023: Senthil Chandrasegaran, Assistant Professor Peter Lloyd, Professor Email: DTM-IO@tudelft.nl	

ID4060	Manage your Master	2
Course Coordinator	Ir. S.H. Gieles	
Contact Hours / Week x/x/x/x		
Education Period	1 3	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Course Contents	MYM introduces students to IDE Community and the structure and possibilities of the MSc program at IDE. It supports students in developing their own, personal vision on their Master education and in developing a plan for the final year of their Master program. MYM takes place on the first three days of the 1st and 2nd semester of their Master program. Each one of these days has a specific theme: Monday in first semester: Welcome to IDE Tuesday in first semester: Your & Our Identity Wednesday in first semester: Opportunities at IDE Monday in second semester: Your 3rd semester Tuesday in first semester: Research Wednesday in second semester: Preparing for graduation Every day provides a combination of activities to build community and help shape your master.	
Study Goals	After following this course, the student is able to 1. Describe what it means for them to be part of IDE Community 2. Formulate their own, personal vision on being a designer 3. Evaluate what they would like to develop further in the field of Industrial Design Engineering 4. Evaluate what are their personal interest areas in the field of Industrial Design Engineering 5. Formulate a plan for developing and filling in their 2nd Master year	
Education Method	The MYM days consist of lectures, group activities, workshops, individual assignments and presentations.	
Literature and Study Materials	Will be provided during the course	
Assessment	During the MYM course you prepare a personal development vision, including a plan for your 3rd and 4th semester. The idea of the vision is that it is a personal presentation and discussion document, in which you present a profile of yourself as Master student (DfI, SPD or IPD), you indicate your designer identity and development ambition, and what you plan to do in order to get there. This deliverable should be uploaded on Brightspace. In the 5 page master plan you display: (1) what characterizes you as a student of your particular master? why did you choose this master? how do you plan to fill it in, in your unique way? (LO1, LO2) (2) what are your current skills? what are you good at? what skills would you like to develop? consider different types of skills: technical, design, research, organizational and people skills. (LO3) (3) what are your dreams in your professional life? what are your passions? how do you see your future? in what kind of organisation would you like to work (profit/non-profit, own company, big/small)? where do you plan to work? how do you prepare for this career? (LO4) After completing the MYM days in the 1st semester, you need to upload the first three pages on Brightspace. (4) the courses you plan to follow in the 3rd semester: electives, Design-XL, internship, research project, external courses, and so on. Also think about how you prepare for these courses. (LO5) (5) your desired topic of your graduation project in the 4th semester, including suggestions for supervisors and involvement of external organisations (e.g., companies); also think about a back-up plan, in case things work out differently (LO5) After completing the MYM days in your 2nd semester, you need to upload all five pages in Brightspace. This time, your MYM plan will be assessed in terms of Pass or Fail. You can submit your master plan twice a year on Brightspace. Submission deadlines are at the end of the third week in Q1 and Q2.	
Enrolment / Application	For IDE MSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	
Remarks	Students should register for the MYM course at the start of the first semester of their Master program. You only need to register once for this course; you do not need to register again in your 2nd or 3rd semester.	

ID4070	IDE Academy	4
Course Coordinator	Ir. A.J.M. van Leeuwen	
Contact Hours / Week x/x/x/x		
Education Period	1 2 3 4	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Course Contents	The IDE Academy is a series of one-day workshops, always on Wednesdays, on design-related skills, knowledge and insights. Each Master's student enrolls in the academy to participate in 14 such workshops (7 workshops for exchange students), rounding off enrollment with a personal reflection. A typical academy workshop provides basic training on a relevant set of transferrable IDE skills. Sample topics relate to research (e.g. observational research, literature surveying), applying methods (e.g. scrum, business models, innovation management), conceptual design (e.g. visualisation skills, generative design), prototyping (e.g. screen-based prototypes, programming), technology (e.g. VR tools, smart materials), and personal skills (e.g. team work, reporting). In most cases, the workshop also provides an entry point for further self-study and development. Industry events, such as business/technology fairs, can also double as workshops: the student actively participates in the event and are given an additional assignment to reports back at the end of the day. The IDE academy makes good use of different networks (alumni, research and practice) by inviting a wide range of design practitioners from industry. This way, IDE Academy builds a bridge between academy and industry.	
Study Goals	- Exploration of diverse skills (14 workshops, 7 for Exchange students) - Recognition of skill improvement - Recognition of the value of skills for their career - Recognition of transferability of the skills learned to other courses	
Education Method	As IDE Academy staff, we encourage students to take responsibility for their own learning, and motivate them to independently seek for new skill development, knowledge, and design insights. We do this by offering around 100 workshops a year over almost 35 weeks. Students attend 14 workshops (7 for exchange students) of their choice. Students can start with The IDE Academy at the beginning of every semester that always kicks off with a workshop where new students can get familiarised with the IDE Academy. In this introductory workshop for both new and current students the workshops for the upcoming semester are presented and they are asked to make a plan for their learning. Every week, students can register (enroll) for a workshop of their choice. Students do not have to do all workshops consecutively. That is not possible anyway because the number of places offered is usually not enough for everyone to participate. They can plan the workshop as they see fit. If enrollment exceeds the maximum of participants, places are randomly assigned. There is a system that guarantees places if students have been rejected too many times. Students are encouraged to complete all workshops of the IDE Academy in one year In most workshops, students actively engage with the subject matter in small groups. This allows for a variety of group work themes to develop. During the workshop, the student's attendance is registered. Immediately after completion, students provide brief feedback about the workshop by means of a quiz and rank it on a 5-point scale from "not useful" to "highly useful". This feedback is then used by the staff to improve the portfolio of workshops and make improvements. Registration of attendance and the quiz are both compulsory to validly complete a workshop. After completing the workshops students are asked to write a personal reflection on their learning by means of a "Letter to Future You". Due to the Master renewal that will be introduced next academic year, IDE Academy students are encouraged to complete the course at the end of this academic year (2023/2024).	
Assessment	Assessment is based on attendance which has to be registered on the Attendance List during the workshop AND by completing the quiz at the end of every workshop. At any time after successfully completing the 14 workshops (7 for exchange students), students have to reflect on the course and the workshops by writing a letter to their future self.	
Enrolment / Application	For IDE MSc (elective) courses registration via the MyTUDelft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See the link below for exact deadlines and details: https://www.tudelft.nl/en/student/ide/practical-affairs/registration-for-courses-my-tudelft Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	

Year	2023/2024
Organization	Industrial Design Engineering
Education	Master Strategic Product Design

Master Specific Courses Strategic Product Design

ID4216	Context and Conceptualisation	6
Course Coordinator	Prof.dr. P.J. Stappers	
Contact Hours / Week x/x/x/x	4/4/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	The Master course 'Context and Conceptualisation' deals with the various contexts surrounding the user, the users' interaction with the product/service, and the system in which that takes place. Examples of such contexts are the situation in which the product/service is used, other people that are involved and the role of the product in the user's life, even when the user is not physically using it. Contexts can be examined from, among others, local, societal, cultural, technological and future perspectives. Over the past decades, there has been an increasing attention for these contexts, and designers have to become sophisticated in (1: academic writing) dealing with scientific knowledge about the context, (2: context research and service design) obtaining situated knowledge about the context from the 'experts of experience', i.e., users, stakeholders, and domain experts, and (3: conceptualisation) devising visions and concept designs, and communicating these with stakeholders. At IDE, methods expertise has been developed in all these three topics, with techniques such as contextmapping, VIP, and various research themes of the faculty. The course presents theory and cases from practice, gives hands-on experience with a number of the techniques, and principles for using the techniques in design projects, and offers training in (academic) writing on a topic connecting the research interests of students and staff.	
Study Goals	On completion of this course, the student will be able to: 1. describe the developments, concepts and methods of context research and conceptualisation; 2. experiment with various design research methods to explore the (current and/or ideal) contexts of use; 3. integrate principles of academic writing.	
Education Method	The course aims to bring both theory and hands-on experience. Further training in these skills is left to MSc projects, such as Exploring Interactions (Dfi), and the Design Strategy Project (SPD). For some elements, e.g., contextmapping, service design, and VIP, there are elective courses in which students can further deepen their skills. For main topics (2: context research and service design) and (3: conceptualisation), lectures bring theory, instruction for practicals, and cases from practice. Practical in groups (which remain fixed for the course) provide experience with techniques, and reflection on the theoretical principles that underly them. Weekly lectures and practicals are often tied in a 'sandwich' construction on a topic, e.g., 'qualitative analysis', or 'empathy'. The opening lecture introduces the topic and relevance, and gives instruction for a self-managed group assignment. In the practical one technique in the topic is tried out and the group reflects on the underlying principles. The closing Q&A deepens the theory and brings cases from industrial practice. Often, assignments continue on the results of the week before. Students also read the book and reader accompanying the course; their knowledge of these is tested in an open-book 'written test' (individual work). Weekly topics include the notion of context, contextmapping techniques, cultural probes and generative tools, analysis techniques for experience design research, visualisation techniques (mindmaps, infographics, storyboards) for communicating user experiences, academic writing and review, and basic service design techniques. The practicals and assignments are meant to provide a first experience of the methods and techniques. For main topic (1: academic writing), IDE researchers provide themes, and students individually select one theme. Two plenary lectures on writing are provided, and in two workshops, the researcher and students develop the students' individual argument and paper. Most of the writing activity is individual, and students get feedback from other students and staff on their draft (in two workshops) and paper (in a blind peer review round).	
Literature and Study Materials	The course is accompanied by (1) a course book giving theory, guidelines, case examples, and experiences of international design researchers. Sanders, E. B. N., & Stappers, P. J. (2012). Convivial toolbox: Generative research for the front end of design. Amsterdam: BIS. (2) a reader with key articles on the topics discussed, Stappers, P. J., & Sleeswijk Visser, F. (2023) Context & Conceptualization Reader. Faculty of Industrial Design Engineering, TU Delft. [only few changes to 2020 and 2022 editions] Items (1) and (2) can be brought in hardcopy or digitally to the written test (the pages do not have to be 'clean' of notes). (3) a separate manual on the academic writing exercise, provided digitally on Brightspace (4) Brightspace is used throughout the course for notifications, lecture sheets, background materials, and handing in	
Assessment	In 2023-2024, assessment is made on three elements: - written test about book and reader (30%, individual) - active participation and handing in work from the practicals (20%, individual) - academic paper (50%, individual)	
Enrolment / Application	For IDE MSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	

ID4315-16	Brand & Product Commercialisation	8
Course Coordinator	Ir. S. Bakker-Wu	
Contact Hours / Week x/x/x/x		
Education Period	3 4	
Start Education	3	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	How to turn your innovative product or service into a commercial success? Designers need to think upfront in the New Product Development (NPD) process about how to introduce the product or service in the market later on. What is the brand strategy? Which groups of consumers to target? What is the pricing strategy? What are the distribution channels? All these, and many other questions, need to be answered in a market introduction plan. In-depth analyses of the company, the customers, the competitors and the context have to be conducted. In this course, different tools, models and research methods will be introduced. To put theory into practice, students are required to write a market introduction strategy (group assignment) for a new product or service.	
Study Goals	Upon completion of this course, you will be able to 1. Understand several brand models, analyse an existing brand using one of them, and use them to create a new one. 2. Analyse product, company, competitors, consumer and context in the light of brand development and use this outcome to create a brand (marketing) strategy for new and existing brands. 3. Understand the determinants of new product success and apply the theory to create a new product launch strategy. 4. Understand the role of the marketing mix, analyse company marketing mix elements and create a marketing mix for new product launches. 5. Apply verbal, textual, visual and multimedia skills to create communication with different stakeholders of a new product, such as customer and company decision-makers.	
Education Method	The students will receive blended learning activities including the following elements - Literature self-study - (Guest) lectures to explain and discuss the key learnings - Group assignment: to apply your knowledge and insights. Literature & study materials: Papers and books will be announced on BrightSpace.	
Assessment	The final grade of the course consists grades of group assignments Information on the assessment criteria and the composition of final grade will be announced on BrightSpace.	
Enrolment / Application	For IDE MSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	

ID4330	New Product Economics	3
Course Coordinator	N. Kyriakopoulos	
Responsible for assignments	Ir. R.G.H. Bluemink	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	This course will focus on the financial side of new product and service developments. A designer mainly focuses on creating meaningful solutions that fulfil customer needs. Although this is obviously crucial, it is just as important that the new product or service is financially viable. No matter whether you want to launch your own start-up, convince potential investors of your idea, or you will be working in a big corporation, and you need to convince the management of your idea, you will need to be able to assess and communicate the financial viability of a project. In this course, you will learn how to strategically and (most importantly!) financially approach the development of a new product or service.	
Study Goals	At the end of this course, students are able to critically and financially evaluate new product opportunities and make economically well-informed NPD decisions. More specifically: 1. Students understand and are able to explain the key concepts of financial management and management accounting 2. Students are able to set up a financial business case for a new product opportunity (including capital needs planning, income statement, cash flow planning and break-even analysis) 3. Students are able to assess the financial viability of new product development projects	
Education Method	Weekly lectures and individual and group assignments.	
Assessment	Examination by an individual digital exam and a group assignment. The final course grade for New Product Economics is compounded by the grade of the individual digital exam and the group assignment grade.	
Enrolment / Application	For IDE MSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	

ID4337	Design Roadmapping	3
Course Coordinator	Dr.ir. W.L. Simonse	
Contact Hours / Week x/x/x/x		
Education Period	1	
Start Education	1	
Exam Period	Different, to be announced	
Course Language	English	
Required for	Strategic Product Design Master curriculum	
Course Contents	Timing is of strategic importance in leveraging on technology inventions and capturing new values embraced by users. Timing is also crucial in anticipating the ultimate moments of launching new products and service propositions. As part of the strategy development process in organizations, design roadmapping is used for developing the innovation strategy. Design Roadmapping devises creative responses to future strategic challenges. It connects user values to future visions and details an evolutionary pacing of design innovations. Mapping value propositions and their versions on the future timeline ensures continuous innovation for your organisation in the long run. From strategy & organization theory and design methodology perspectives, this master course offers a learning experience of strategic designing long term foresights. Besides the concepts and definitions related to the theoretical framing of design roadmapping, the latest research results and case examples from the experience of design and strategic practitioners are shared. The aim of the course is to create future visions for the destination and horizons of a strategic roadmap - and craft a tactical roadmap - with new propositions in an evolutionary pace- in order to support strategic choices and cross-functional communication. Guided by the future foresight techniques of design roadmapping, you will learn how to uncover new trends, scout for new technologies and map the values and ideas on the roadmap. The models and methods are organized around a process of diverging and converging activities: - Future visioning and creative trend research - Mapping horizons of design innovations - Technology scouting and idea mapping - Visualizing the roadmaps During the course you will learn how to apply these models and methods in mapping design innovations for the project assignment. Through strong visualization, you will design a strategic roadmap that supports an organizational mindset on user centred design innovations and a tactical roadmap on innovating new value propositions and versions that lay out the groundwork for turning a future vision into a reality.	
Study Goals	The Design Roadmapping learning objectives are: 1. Learn to frame and analyze strategic design practice with roadmapping concepts and frameworks from the literature. 2. Learn to create a future vision within the dynamics surrounding the strategic development of an innovation strategy. 3. Learn to formulate a time pacing strategy with respect to life cycle evolutions and constraints in resources. 4. Learn how to apply creative trend research to formulate future visions. 5. Learn to map a tactical roadmap with strategic modules and scouting for new technologies. 6. Learn to visualize roadmaps that synchronize user values, product/market propositions and technology innovations to support cross-functional communication between different stakeholders.	
Education Method	The education method is roughly inspired on the experimental learning approach blended with digital learning experiences. It entails that every week you read a chapter of the course book and watch a video lecture on a self-study base, after which you meet up with the lecturer and peers in an online session. In these online meetups we discuss: how to apply the models and methods into strategic design practice. To stir up these conversations, for each meetup one of the teams will introduce an example of application with a video lecture on a case. An essential part of the education method is that we work in a real life collaboration context of experimental learning with a public or private partner. The partner offers a real life problem relevant to the learning objectives of designing roadmapping. The learning outcomes of this real life project assignment serve as an examination part of the course. To practice your strategic design skills there are self-paced group exercise provided online and related to the Lab in the course book. With this group exercise you can check your readiness for the real life application on the project assignment.	
Literature and Study Materials	Study book: Design Roadmapping by Lianne Simonse. Amsterdam, BIS Publishers.ISBN 978-90-6369-459-3. Additional compilation of articles, videos and links will be provided on Brightspace.	
Books	Study book: Design Roadmapping by Lianne Simonse. Amsterdam, BIS Publishers.ISBN,978-90-6369-459-3.	
Reader	Additional compilation of articles, videos and links will be provided on Brightspace	
Assessment	A Design director of an organization will join the course kick-off and end presentations of the design roadmapping course. A tailored project assignment is part of the assessment. The examination of the course consists of two team assignments: the lecture assignment and the project assignment. The deliverable of the lecture assignment is a short presentation (slides and oral presentation). The deliverable of the project assignment consists of visual roadmaps and argumentation report. The lecture deliverable accounts for 30% , and the project deliverable accounts for 70% of the end mark. The end mark calculation is rounded on ,5 and ,0 marks.	
Enrolment / Application	For IDE MSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	

Special Information	Course coordinator: Dr.ir. Lianne W.L. Simonse. e-Mail: L.W.L.Simonse@tudelft.nl
Remarks	This is a collaboration course. External partners bring in real life cases and obtain IP rights to the results. At the start of the course the course staff will ask you to sign the IP(R) transfer with TU Delft to make collaboration possible. More detailed information can be found in the student portal -> practical affairs -> Intellectual Property in IDE education -> collaboration courses.

ID4340	Strategic Value of Design	3
Course Coordinator	G. Calabretta	
Contact Hours / Week x/x/x/x		
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Strategic Value of Design (SVD) is a reflection course that enables you to develop and argument an informed perspective on how design can create value for companies and for the society. This perspective will be discussed and challenged during the course through, and finally embedded in your future oriented vision for strategic design and your individual positioning as a design professional in the innovation industry. As strategic design has achieved a certain degree of maturity and there is a better understanding of its role in innovation, we want to integrate a future perspective in the discussion. Thus, the course will combine thoughts and perspective on the role of strategic design nowadays and in 2030. Will strategic designers become CEOs? Or will strategic design disappear? What is the impact of the current crisis on the future of the strategic design industry?	
Study Goals	At the end of the course students will be able to: 1. Summarise and compare historic and current developments in theories about the strategic value of design and designers skills and capabilities 2. Develop, argument and defend their own opinion on the strategic value of design and designers skills and capabilities 3. Formulate a positioning statement and create a related tool/method that embed designers distinctive skills and capabilities compared to other designers and non-designers working in the area of innovation. 4. Demonstrate the ability to use their visualisation and story-telling skills to communicate to a professional audience (e.g., clients, potential employers)	
Education Method	The course will use a variety of learning activities to support you in your future-oriented reflection, including lectures, discussions, coaching sessions, and reflection moments.	
Literature and Study Materials	- Online video tutorials - Articles (to be announced)	
Assessment	- Individual assignment 1: writing a critical summary about a strategic design topic - Individual assignment 2: positioning yourself as a design professional in the market	
Enrolment / Application	For IDE MSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	

ID4350-16	Design Strategy Project	12
Course Coordinator	R.A. Price	
Contact Hours / Week x/x/x/x		
Education Period	1 2	
Start Education	1	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	*The following course contents and description may be subject to change due to Covid-19* This course is a combination of background information on Design Strategy as well as practical assignments for companies. In Q1, module work will acquaint the students with strategic design. In Q2 students will work with a companies on a strategic design project. The cases range in different areas of design strategy going from identifying new market opportunities, developing new innovation approaches, investigating how the company can leverage their competences to proposing new visioning projects and disrupting existing organisational norms. Topics covered during the interactive lectures of the course are for example: innovation strategies, dealing with digital innovation and technology, developing capabilities in an organisation, designing for implementing your strategic advice and supporting organisational change as a designer. The lectures and the assignments are closely related to each other. Example of the companies we worked with: Brabantia (2003-2004) Johnson Controls (2004-2005) Oce (2005-2006) Mexx (2006-2007) Reinier de Graaff groep (2006-2007) Philips (2007-2008) TomTom (2008-2009) DSM (2009-2010) The Schiphol Group (2010-2011) VDL (2011-2012) Daimler-Benz (2012-2013) Rabobank (2013-2014) Eneco (2014-2015) KLM (2015-2016) Adidas - Ford - KLM (2016-2017) Ford - BARCO - Vanderlande - Werkspot (2017-2018) Ford - Vanderlande - Barco - KLM (2018-2019) Ford - Netherlands Red Cross - Barco - Yanfeng (2019-2020) Ford _ Netherlands Red Cross - Barco - TU Delft (2020-2021)	
Study Goals	1) Analyse products/r services and organisations using a strategic lens 2) Reframe identified problems into strategic design opportunities 3) Design new product/s, service/s and system/s that allow an organisation to grow with strategic intent 4) Develop design management skills to manage projects with external stakeholders	
Education Method	The education methods are:. 1) Lectures consist of theoretical and practical content. 2) Project work completed in teams allows students to implement lecture content in a practical way - coached weekly to ensure structured learning and reflection points. 3) Readings allow students to enrich their learning	
Assessment	The students will be evaluated in the following ways: - formative and summative presentation (and materials) of Module 1 and 2 - formative and summative presentation (and materials) of a Client Design Strategy Assignment	
Exam Hours	No exam	
Enrolment / Application	For IDE MSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	
Remarks	This is a collaboration course. External partners bring in real life cases and obtain IP rights to the results. At the start of the course the course staff will ask you to sign the IP(R) transfer with TU Delft to make collaboration possible. More detailed information can be found in the student portal -> practical affairs -> Intellectual Property in IDE education -> collaboration courses.	

ID4355-18	SPD Research	13
Course Coordinator	Dr. M.E.H. Creusen	
Contact Hours / Week x/x/x/x		
Education Period	3 4	
Start Education	3	
Exam Period	3 4 5	
Course Language	English	
Course Contents	The Master course SPD Research provides students with knowledge about qualitative and quantitative research methods, focused both on scientific research and on customer research within the practice of NPD. In addition, students learn to carry out a scientific research project in the area of Strategic Product Design. The course consists out of a theoretical and a practical part and has a total workload of 13 EC. The four different modules in this course are: Practical part (50% of the course end grade): SPD Research Project (SPD-RP) Theory part (together 50% of the course end grade): SPD Research Qualitative Methodology (SPD-QL) SPD Research Quantitative Methodology (SPD-QNT) Customer Research Methods (CRM) The SPD-QL and SPD-QNT modules (quarter 3) respectively provide students with knowledge about qualitative and quantitative research methodology. In the SPD-RP module (one semester), students gain further experience in either qualitative or quantitative research (their choice) by performing a scientific research project in groups of 3, coached by a DOS faculty member. The CRM module (quarter 4) focuses on customer research methods that are used in the practice of New Product Development. These four modules are described in a bit more detail below. SPD Research Project (SPD-RP) Within one semester (Q3 and Q4) students learn how to design, carry out, analyze and report a scientific research project in the field of strategic product design, using either qualitative or quantitative methods. In groups of 3, students participate in a faculty members research topic. The needed (qualitative and quantitative) methodology knowledge for this research project is provided in the SPD-QL and SPD-QNT modules (see below). SPD Research Qualitative Methodology (SPD-QL) This module (Q3) teaches students qualitative research techniques in relation to Strategic Product Design. Students practice with applying the theory via exercises, which are discussed in interactive sessions. This module supports students in applying and understanding scientific research methods so they can (1) successfully perform qualitative research projects and (2) critically evaluate the relevance and value of outcomes of qualitative research. SPD Research Quantitative Methodology (SPD-QNT) This module (Q3) teaches students quantitative research techniques in relation to Strategic Product Design. Practice with the analysis of quantitative data is provided in tutorial sessions. This module supports students in applying and understanding scientific research methods so they can (1) successfully perform quantitative research projects and (2) critically evaluate the relevance and value of quantitative research outcomes. Customer Research Methods (CRM) This module (Q4) introduces methods used in practice for collecting input from customers/consumers to support the first stages of the product development process. Both qualitative and quantitative methods (and combinations) are treated (e.g., Lead User Analysis, Concept Testing). This module provides students with basic knowledge helping them in performing customer research themselves and in critically evaluating the relevance and value of customer research outcomes.	
Study Goals	On completion of the course SPD Research, students will be able to: SPD Research Project (SPD-RP) 1.1 formulate and justify a specific research question based on the topic set by the supervisor 1.2 set up and argue for a research design to conduct an empirical study for investigating the research question 1.3 conduct an empirical study, analyze the results, and integrate and evaluate the outcomes of the study 1.4 write up the project (theory, study design, analysis and results) in a scientific article and poster SPD Qualitative Research Methodology (SPD-QL) 2.1: Evaluate the suitability of different qualitative research designs by: 2.1a Describing and comparing different qualitative data collection and analysis methods 2.1b Identifying their strengths and weaknesses, and discussing their suitability for investigating different types of research questions. 2.2: Evaluate and improve the quality of qualitative research by: 2.2a Explaining the academic criteria for assessing research quality, and examining the quality of research using these criteria. 2.2b Proposing strategies to enhance research quality. 2.3: Plan, practice and improve qualitative data collection and analysis by: 2.3a: Developing and improving a semi-structured interview guide. 2.3b: Analysing interview data using a grounded theory approach. SPD Quantitative Research Methodology (SPD-QNT) 3.1 describe and compare different quantitative data collection and analysis methods, identify their strengths and weaknesses, and understand their suitability for investigating different types of research questions 3.2 recognize and explain the academic criteria for assessing research quality, and appraise the quality of existing quantitative research using these criteria 3.3 analyze quantitative data, interpret and report analysis results Customer Research Methods (CRM) 3.1 describe and compare the different consumer research (CR) methods treated in the module and understand when and how to use each of them 3.2 understand and explain the strengths and weaknesses of these different CR methods 3.3 select an appropriate CR method for a particular CR need during the new product development process and justify the choice	
Education Method	In the SPD Research Project, each student group receives regular coaching by their coach/supervisor, and participates in three presentation sessions with the other students in the topic. In addition, there are lectures about literature search and scientific article writing.	

	In the SPD-QL module, students gain knowledge about qualitative research techniques through articles in a reader, and practice with setting up, evaluating and improving qualitative research via exercises that are discussed in interactive sessions. In the SPD-QNT module, lectures about quantitative methods are followed by (computer) tutorials in which students practice quantitative analysis methods with experts present. In the CRM module, lectures about customer research methods used in product development and guest lectures about the practice of market research are given.
Literature and Study Materials	For the SPD-RP module some relevant starting literature is provided by the topic supervisor and students should gather other relevant literature themselves. In the theory modules within the SPD Research course readers with articles/book chapters (SPD-QL, CRM) or handouts (SPD-QNT) are used, and lecture slides are made available on Brightspace.
Assessment	The course end grade is determined for 50% by the practical part (SPD-RP) and for 50% by the theory part. The grade for the theory part is the average of the grades for the three theory modules (SPD-QL, SPD-QNT, CRM). The grade for the project part (SPD-RP) has to be at least a 6.0 in order to pass the SPD Research course. Grades for each of the theory modules (SPD-QL, SPD-QNT, CRM) have to be at least a 5.0 in order to pass the SPD Research course. (NB, when you do not pass the whole course in one year, only module grades of 6.0 or higher are still valid in the next year, see the TER of IDE). The SPD-RP module grade is based on an academic research article and accompanying poster. There are three formative subparts that are presented to the other students in the topic and coach feedback is given on a written part deliverable, useful for improving the part for the final article. The grade for the SPD-QL, SPD-QNT and CRM modules is determined by means of a written exam for each of them. The modules and their components are graded between 1.0 and 10.0 and not rounded off. The end grade of the SPD Research course is rounded off to whole or half grade points (i.e., 6.0, 6.5, 7.0, etc.) and has to be at least a 6.0 in order to pass the course.
Enrolment / Application	For IDE MSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details.
Special Information	Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty. Course email: ID4355-io@tudelft.nl SPD Research course coordinator: Dr. Mariëlle Creusen, room B-4-070

ID4370	SPD Media	3
Course Coordinator	M. Guerreiro Goncalves	
Contact Hours / Week x/x/x/x		
Education Period	4	
Start Education	4	
Exam Period	Different, to be announced	
Course Language	English	
Expected prior knowledge	Pre-requisites for SPD media There are no specific knowledge prerequisites to follow SPD media. You are able to select one media skill (from the available four or five media tracks) and follow 6 workshops on the chosen one. Each students entry level will invariably be different from the others, with people having some experience to no experience within each media skill. Thus, you do not need to know already how to make a video or to sketch to be able to follow SPD media you will have the opportunity to follow workshops and to individually shape your progress. Nevertheless, you should have previously followed design courses and have some projects in your portfolio you wish to improve. Regardless, there are some workshops within SPD Media that have pre-requisites, limiting the entry of students in some workshops if they are already proficient in them (as an example, in some of the media tracks, it is important to be acquainted with Adobe Illustrator). Check the manual of the course for more information about specific pre-requisites.	
Course Contents	This is a master mandatory course, which aims to support SPD master students in visually expressing their strategic skills. In the SPD master programme, you have followed courses that teach you how to think and design strategically. But as students in an Industrial Design Engineering faculty, besides designing in a strategic way, you are also expected to communicate your work in a clear and appealing manner. The SPD media course is the platform for you to practice skills that will enable you to create and communicate engaging and effective material, in order to enhance how your work is perceived, by colleagues, clients and other stakeholders. You will also learn and practice core concepts of storytelling and gestalt principles (such as spacing, proportion, and balance). In SPD media you are able to choose one media skill, from a number of media tracks (e.g., Sketching; Graphic Design; Animation; Data Visualisation and Auditory Storytelling). You will receive individual support on one of these media skills and you will have the opportunity to tailor your education with the expressive means you find most important. These skills will be useful in the coming years of your academic and professional life, as you position yourself as a strategic designer, with unique (visual) communication abilities. This course has a total of 3EC, which results into 84 hours of individual work for the students (in and outside of class).	
Study Goals	After SPD Media, you will be able to: LO1: Apply and illustrate core basics of gestalt theory, storytelling and visual communication in media skills artefacts. LO2: Apply media skills to better express and showcase a variety of SPD-related knowledge and capabilities. LO3: Create a final project showcasing your portfolio of media skills, developed during the course. LO4: Evaluate and reflect on your progress (compared to your starting point and the end result).	
Education Method	Course Philosophy This course applies a project-oriented learning basis, at the individual level. This means that you have the opportunity to learn and apply the media skills (for this course and for your portfolio), in order to improve your expressiveness. SPD media is organised in three components: lectures; workshops; and the showing of your work (in the final exhibition and other in-course events). You will learn and practice your chosen media skill, using a hands-on approach, directly working (individually) on any project you wish to improve. You will have the possibility to actively explore, in an independent manner, your own preference of expressive means, supported by the coaches and experts of this course. The teaching methods of SPD media are the following: - A kick-off lecture, to get to know the media skills available, talk with coaches, and figure out which media tracks is the best for you; - Mini video-lectures with learning activities; - Reflection on your progress, development and personal learning objectives. - Media skills workshops with learning activities; - Individual assignment, where you can apply insights from lectures and feedback from the workshop coaches and colleagues. - Final online exhibition of your work.	
Assessment	The final grade is composed by two elements - Your media track project (85% grade); - Completion of three reflection boards (15% grade). Your media track project is assessed individually, based on specific rubrics for each of the media tracks (in relation to the project).	
Enrolment / Application	For IDE MSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	

Year	2023/2024
Organization	Industrial Design Engineering
Education	Master Strategic Product Design

2nd year Master Strategic Product Design

ID4390-16	Graduation Project (SPD)	30
Course Coordinator	G. Calabretta	
Contact Hours / Week x/x/x/x		
Education Period	None (Self Study)	
Start Education	1 2 3 4 5	
Exam Period	none	
Course Language	English	
Course Contents	The graduation project gives the student the opportunity to show he is worthy of the Master of Science title and that he fulfils the requirements of the program. This implies that the student uses his knowledge and skills as an independent industrial design engineer to execute a complex project. The graduation project can be seen both as the masterpiece of the student, and as a (personal) learning experience. The scope is therefore not only on testing competences, but also on the development of knowledge, understanding and skills of the student during the project. The graduation project is considered to be a stepping stone to a future professional career. A high level of independence is therefore expected from the student in the planning and execution of the assignment and in the acquisition of knowledge, understanding and skills. A student may start the graduation project under the condition that he has passed all mandatory/first year courses of the Masters Program and the Project Brief is approved by the Board of Examiners.	
Study Goals	On completion of this course, the student will be able to: - effectively collect, analyse, generate and evaluate knowledge required for the project - justify his/her choices with respect to used methods and/or approaches used in the project. - deliver a relevant (feasible, viable and desirable) project result - effectively and thoroughly communicate to- and discuss with stakeholders involved in the project - manage a design/research project independently within the given time	
Assessment	During the graduation project, there are three formal moment when the students work is assessed; Mid-term, Green-light and final assessment. The assessment of the graduation projects is supported by means of a rubric. The deliverables for the graduation project consist of; - A report providing insight in the process and project results, - A showcase; either a prototype or model, a movie or a poster - A public presentation (at the graduation day) An evaluation of the project with the supervising team, consisting of a chairman and a mentor from IDE and, when applicable, an external mentor is scheduled directly following the public presentation.	
Remarks	A more elaborate description of the graduation project can be found in the document Graduation Manual. (See; http://io.studenten.tudelft.nl/ - education - Graduation and Internship - Graduation (MSc)- Download - Graduation manual for students)	

Year	2023/2024
Organization	Industrial Design Engineering
Education	Master Strategic Product Design

Electives (30 EC) SPD

Year	2023/2024
Organization	Industrial Design Engineering
Education	Master Strategic Product Design

Specialization Medesign

Year	2023/2024
Organization	Industrial Design Engineering
Education	Master Strategic Product Design

Compulsory Electives (min. 9 ECTS) Medesign
--

ID5301	Biomechanics	3
Course Coordinator	Dr. T. Huysmans	
Contact Hours / Week x/x/x/x		
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Digital human models (DHM) are computer representations of humans that allow to simulate the interaction of the digital human with a virtual product or environment. Since their introduction in the 1960s, enormous progress has been made but it is still a very active research area due to the complex nature of the human body. A variety of DHM software tools are currently available to efficiently incorporate human factors early-on in the design process and they are used to improve function, fit, and comfort of products in e.g. automotive, aviation, defense, fashion, sports, and medical industries. This course focusses on two emerging DHM areas: 3D anthropometry and biomechanical modelling and simulation, with a focus on the musculoskeletal system (MSK). The course will start with an overview of the function of bones, muscle, and tendons of human body and introduce basic biomechanical concepts. In a flipped classroom setting, students will familiarize themselves with the range of available DHM technologies and tools. Through example cases and a group assignment, students will learn to analyze (dynamic) human-product interactions, how to optimize them using DHMs, and finally how to set up a physical validation study to validate design choices. Example cases include optimizing bicycle seat height and improving respiratory mask fit. Typical products that are treated are tools, supports (chairs and beds), shoes, wearables, and medical equipment.	
Study Goals	After completing the Digital Human Modeling course, students should be able to: LO 1: Explain the general biomechanical functions of bone, muscles, ligaments, tendons, and joints of the human body; LO 2: Apply simple biomechanical modelling techniques on paper; LO 3: Classify digital human modeling (DHM) tools according to purpose and underlying technology; LO 4: Analyze human-product interactions and identify critical body dimensions and product parameters that affect the interaction; LO 5: Employ existing 3D anthropometry and musculoskeletal (MSK) simulation software to analyze fit and biomechanics of simple human-product interactions; LO 6: Propose product modifications to improve product performance in terms of function, fit, safety and comfort; LO 7: Identify an approach (hardware, software, criteria) for evaluation of fit and biomechanics in practical human-product interactions.	
Education Method	lectures, flipped classroom, workshop, and peer instruction.	
Literature and Study Materials	Reader, course slides, scientific articles, DHM software.	
Assessment	group assignments	
Enrolment / Application	For IDE MSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	

ID5383	Inclusive Design	3
Course Coordinator	Dr. S.U. Boess	
Contact Hours / Week x/x/x/x		
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The student has learned the design relevant terms and organisational structures in the world of inclusive design (and how it is different from and similar to design for all, or universal design). The student has practiced inclusive design in a case study (group work) for a specified population as mentioned. The student has involved a few persons of her or his target group in the assignment.	
Study Goals	The main course goal is to understand the current state of the phenomenon of Inclusive Design, which had alternative names like Design for All or Universal Design. Inclusive Design means to aim for the widest possible audience during and with the design process. It does not mean however to design for all 7 billion people on earth! It means considering any specified population to include in designing, and how solutions could also benefit other groups or a wider group. For example: left-handed people, wheelchair users, tall people, small people, obese people, colour blind people, people with dyslexia or RSI, people with age-related specifics etc. The list continues, which means that we realise that hardly anyone is 'normal'. People aged over 65 for example, currently count 15% in the Netherlands but this will grow to 25% in 2050. They are more diverse than younger people. Risks for them also rise: lethal home accidents rise 5-fold compared to younger people. The effect of design should be that participation and quality of life is enhanced, and stigmatisation and risk is avoided. Take the example of eye glasses: they were an orthosis in the middle ages and are now aesthetic lifestyle enhancements. To design inclusively, the designer needs to be able to collaborate with people, understand understand their usage of their everyday things, and respond creatively.	
Education Method	(Guest) lectures in combination with group work on design for and with specified populations.	
Literature and Study Materials	Simon Keates & John Clarkson (2004): Countering design exclusion, an introduction to inclusive design. Springer. Websites: http://www.inclusivedesigntoolkit.com and http://www.hhrc.rca.ac.uk/	
Assessment	The assessment is based on - course participation - team oral presentation and final report - a short written individual essay.	
Enrolment / Application	For IDE MSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	
Special Information	s.u.boess@tudelft.nl	

ID5663	Contextmapping Skills	6
Course Coordinator	Prof.dr. P.J. Stappers	
Contact Hours / Week x/x/x/x		
Education Period	1 2	
Start Education	1	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Participation in 'Contextmapping Skills' is open to all MSc students who believe that end-users are important sources of information for leading new product and service development. Because the elective builds upon the theory and practice taught in ID4216 ('C&C' required for Dfl and SPD), following that course prior to, or in parallel, is highly recommended. Participants are asked to supply a short written motivation and (Manage your Master) portfolio about his or her experience with user research (to both P.J.Stappers@TUDelft.nl and Benjamin@Muzus.nl) to help us prepare the course. Furthermore, we highly recommend you to plan enough time for this course (at least 8 hours for groupwork, including fieldwork, available per week).	
Course Contents	Contextmapping is a design (research) method for including users and other stakeholders in a design process as 'expert of their experience'. By applying generative techniques in the research process, underlying needs, opportunities, motivations and dreams can be discovered. Over the past two decades, the method and its underlying principles and theories have been developed at TU Delft in close collaboration with Dutch and international researchers and practitioners. This development has been done in about a dozen PhD projects, hundreds of MSc projects, and two (online) masterclasses, most of these in close collaboration with industry (see http://studiolab.ide.tudelft.nl/studiolab/contextmapping/resources/). The Master elective course 'Contextmapping Skills' gives students in-depth experience and a chance to develop their skills at these techniques. It is the best opportunity to get specific training from the experts on the method, in a project with a real-world brief and client. The course has a hands-on character, combining elements of reviewing the state of the art, scientific principles, and exploring and practising methods of - determining focus and scope - recruiting participants - designing sensitising materials and workshop materials - setting up, conducting generative group sessions or interviews - analysing field data - creating concept designs grounded in use insights - communicating the insights and designs to stakeholders and clients.	
Study Goals	Upon completion of the course, the student will be able to - understand theories underlying contextmapping techniques - understand, select, and evaluate techniques for a project - apply the techniques in the field - extend the techniques for a specific target group or situation	
Education Method	The course covers both theory and practice. Both are taught in close collaboration between TU staff (http://studiolab.ide.tudelft.nl/studiolab/contextmapping/) and service design agency Muzus (http://www.muzus.nl/en/). The theory part consists of lectures and case examples. In the practical part design research techniques are applied by the students in a project for a 'real' client. In small groups, students will perform the whole contextmapping process in close collaboration with the client, user and tutors. Results will be implemented by the company, helping them towards user-centred innovation.	
Assessment	Assessment is done on the basis of work performed during the course. In part this is judged on the basis of the contribution to the research elements, in part a portfolio / final report and a presentation / workshop at the client is assessed. The precise method of assessment is made known at the first meeting of the group.	
Enrolment / Application	For IDE MSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	

ID5721	Capita Selecta Medisign	3
Course Coordinator	Dr.ir. M. Melles	
Contact Hours / Week x/x/x/x	-	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	The field of design for healthcare is broad and diverse. As a medical design engineer you will work in different positions with professionals from various backgrounds. In this course, the broadness of the field of healthcare design is explored by means of a series of guest lectures of healthcare designers, medical professionals, Medisign alumni, healthcare design researchers and Medisign graduates. Medisign students will become acquainted with different perspectives on design for health and have the opportunity to build their professional network. This course encourages an exchange between professionals and Medisign students during the lectures at which design challenges and research topics in the medical field are presented and discussed. The course will prepare students for their graduation in healthcare; they will learn about the different domains in healthcare, learn how and where design can play a role in healthcare, and learn what their role as designer can be. Following this course is highly recommended for all students following the Medisign specialisation.	
Study Goals	The main objective of this course is to get acquainted with the broad landscape of design for healthcare and to bring students into contact with professionals who are in different ways involved in the development of products or services for healthcare. This way the student is encouraged to reflect on their medical design skills and preferences for a future career. At the end of this course the student should be able to differentiate between the different roles a medical design engineer can have and recognise the different perspectives of healthcare professionals, engineers, and researchers on medical product development.	
Education Method	The course consists of a series of guest lectures for which guests are invited from different areas within the field of medical design (e.g., healthcare designers, medical specialists, Medisign graduates, medical researchers from various backgrounds). Next to this, students need to work on a short design assignment to explore the possibilities for a Medisign graduation project linked to a patient and/or health professional. Guest lecturers are often interested in initiating a graduation project or offer an internship. Capita Selecta encourages and facilitates these opportunities.	
Literature and Study Materials	Study materials will be available via Brightspace.	
Assessment	Students need to attend all 7 lectures, actively take part in the discussions, and carry out a short design assignment. The assessment of this course is formative.	
Enrolment / Application	For IDE MSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	
Special Information	Dr.ir. Marijke Melles E-mail: m.melles@tudelft.nl	

ID5732	Health Psychology: Tools and Methods	3
Course Coordinator	Dr. J.J. Kraal	
Contact Hours / Week x/x/x/x		
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Course Contents	When doing design research designers often encounter psychological topics. Especially in healthcare related projects, topics such as motivation during rehabilitation, coping with stressful situations, and medication adherence require understanding of psychological theory and methodology to come to a good design. Yet psychological methods (including questionnaires and patient-reported outcome measures) and models are not commonly embedded in the design process. Using validated questionnaires will help designers to execute high quality research and improve the reliability of their outcomes. Furthermore, knowledge of psychological models will help designers understand which psychological factors influence behavior and will inspire and guide them in their design process. This course provides insight into the content and applicability of psychological models & methods for design research in the healthcare industry. The content will be taught through a series of theoretical lectures and practical assignments that serve as input and deeper thought about the lecture content. Parallel to the lectures, students will work on a group design project under supervision of a design coach.	
Study Goals	LO1: Describe the basic elements (e.g. concepts, variables) in psychological theory. LO2: Explain how individual differences and the social environment influence health using the psychological theories covered in the lectures. LO3: Identify and analyse at least one example of a design that has used psychological theory in the design process (e.g. definition, ideation, testing). LO4: (Re)design a product, service, or product-service system by applying models and theories from health psychology. LO5: Write and justify a research proposal to evaluate a theory-based design aimed at preventing, improving, or maintaining health.	
Education Method	The students follow lectures (2 hours a week). A design project is interwoven throughout the course and used to apply theoretical and methodological insights. Students carry out the design project as a group but also submit an individual reflection. The design project takes approximately 6 hours per student per week.	
Literature and Study Materials	Journal articles and selected book chapters.	
Assessment	The students' work is assessed on the following items: Research proposal Description and explanation of the health phenomenon, analyses of design examples, and specification of methodology Design process: Theoretical argumentation of design choices, consistency of design with context, and credibility of the design. Reflection: Structured critical reflection of the strengths and weaknesses of the selected theory, methodology and design	
Enrolment / Application	For IDE MSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	

ID5741-23	eHealth Design	3
Course Coordinator	Dr. V.T. Visch	
Contact Hours / Week x/x/x/x		
Education Period	3	
Start Education	3	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	This master elective course focuses on the developments within the eHealth sector and enables students to gain more knowledge about this growing area for healthcare design. Lectures will be provided by the members of the eHealth research group within the IDE faculty. Course topics run parallel to the eHealth research topics at our faculty including user profiles towards health, health and social economic positions, motivation and health, health journeys, health and game design, health and AI interaction, health feedback design, and remote care. In the course you will analyse and present a concept for an improved redesign for an existing eHealth app. These apps will be provided by eHealth developers who will give guest presentations and feedback to your design during the course.	
Study Goals	LO1: The students are able to explain what eHealth is from an industrial design perspective. LO2: The students are able to critically analyze eHealth products in practice based on scientific knowledge. LO3: The students are able to create a concept for a redesign of an existing eHealth product. LO4: The students are able to recommend their eHealth redesign based on theoretical- and practice-based arguments.	
Education Method	The students follow lectures and work in a team of 2 on an eHealth design assignments throughout the course. Every week a new topic is introduced by research staff and guest lecturers of eHealth developers. Knowledge gained during these different lectures have to be applied while working on the assignment. Each lecture is followed by a coaching session hosted by the lecturer. Sharing and discussing insights and knowledge with fellow students, tutors, and eHealth developers (e.g. gathered in desktop research) take a central place in this course. Intermediate and final results are presented to the stakeholders, tutors and fellow students.	
Literature and Study Materials	Literature will be provided during each lecture. With the weekly announcements of the lectures (1 week in advance, through Brightspace) the student can also be asked to go through 1 or 2 relevant papers before attending the lecture. Literature includes several scientific articles or book chapters related to the four different topics. Additionally, up-to-date websites and examples of e-Health applications are provided.	
Assessment	The students' work is assessed on the following two items: Result: Quality and depth of the content of the presentation, report and redesign, capability of linking lecture knowledge to practice. For the report, a template is provided. Communication / presentation: Quality of execution of presentation, showing interest and motivation. Personal communication ability, presenting convincing arguments for own eHealth design, involvement and attitude during joint group discussion.	
Enrolment / Application	For IDE MSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	

ID5752	Rules and Regulations for Designing Medical Devices	3
Course Coordinator	Dr.ir. S.N. Paus-Buzink	
Contact Hours / Week x/x/x/x		
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	When it comes to the approval of a new healthcare product onto the market, safety is one of the most important drivers for success. The ecosystem of stakeholders that are responsible for the overall safety regarding a healthcare product is large and complex. Regulatory authorities control the quality and safety of patients and healthcare professionals is safeguarded for newly developed medical devices. This elective provides insights into the medical design directives regarding safety of healthcare products by focusing on the CE marking process (including ISO norms and risk management). The course includes several lectures of TU Delft tutors and external professional(s) working in the healthcare field. Next to the lectures, students will do a duo-assignments in which they apply insights into practice with the use of a case-study. The course shows students the importance of risk analysis of healthcare products and helps them to integrate this in their design process.	
Study Goals	After completing this course you will be able to: LO1: Interpret relevant procedures, rules and regulations on safeguarding quality and safety of products for healthcare. Identify which are applicable Know how to read and apply LO2: Model the ecosystem of stakeholders and their role(s) in the development and approval process of a healthcare product. Identify which are applicable Understand & explain roles LO3: Analyse and manage risks in medical device development. Conduct risk analysis Apply design interventions to minimize residual risks LO4: Communicate in writing and verbally about conformity of the design and design process with European Regulations Compile the required technical documentation Draw up a declaration of conformity	
Education Method	The course includes weekly lectures and workshops, some given by external experts. Next to some self-study of the provided materials you will practice in applying your newly gained knowledge and skills in a duo-assignment. In this duo-assignment, you will work together with one other student on a case study on a medical product-service system. The overall assignment and sub-assignments are interwoven with the lectures and workshops throughout the course. Some examples of sub-assignments are: defining medical device classification, stakeholder mapping, compiling user- and use-environment descriptions, and conducting use-related risk analysis. During the course, you will provide and receive peer feedback on your sub-assignments. Halfway the course there is also an opportunity to receive formative feedback on your work on the case study from the coaches. At the end of the course, you will give a presentation to the group and submit a report about your duo-assignment.	
Literature and Study Materials	Relevant literature will be provided via Brightspace.	
Assessment	During the course you will receive formative feedback on part of your work on the case-study/duo-assignment from peers and one of the coaches. Summative assessment and grading will be based on your participation in the course activities, final report, presentation and communication with the coaches on your case-study/duo-assignment.	
Enrolment / Application	For IDE MSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	

Ir. S. Bakker-Wu

Unit	Industrieel Ontwerpen
Department	Marketing en Consumentenonderzoek
Telephone	+31 15 27 85359
Room	32.B-4-190

Ir. R.G.H. Bluemink

Unit	Industrieel Ontwerpen
Department	Methodologie en Organisatie van Design
Telephone	+31 15 27 87273
Room	32.B-4-260

Dr. S.U. Boess

Unit	Industrieel Ontwerpen
Department	Applied Ergonomics and Design
Telephone	+31 15 27 83196
Room	32.B-2-140

G. Calabretta

Unit	Industrieel Ontwerpen
Department	Marketing en Consumentenonderzoek
Telephone	+31 15 27 89443
Room	32.B-4-050

Dr. R.S.K. Chandrasegaran

Department	Methodologie en Organisatie van Design
Room	32.B-4-140

Dr. M.E.H. Creusen

Unit	Industrieel Ontwerpen
Department	Marketing en Consumentenonderzoek
Telephone	+31 15 27 85566
Room	32.B-4-070

Ir. S.H. Gieles

Department	Design Aesthetics
-------------------	-------------------

M. Guerreiro Goncalves

Unit	Industrieel Ontwerpen
Department	Methodologie en Organisatie van Design
Telephone	+31 15 27 81311
Room	32.B-4-100

Dr. T. Huysmans

Unit	Industrieel Ontwerpen
Department	Applied Ergonomics and Design
Telephone	+31 15 27 81198

Dr. J.J. Kraal

Department	Applied Ergonomics and Design
Telephone	+31 15 27 81747

N. Kyriakopoulos

Department	Marketing en Consumentenonderzoek
-------------------	-----------------------------------

Ir. A.J.M. van Leeuwen

Department	Human Information Communication Design
-------------------	--

Dr.ir. M. Melles

Unit	Industrieel Ontwerpen
Department	Applied Ergonomics and Design

Telephone	+31 15 27 87514
------------------	-----------------

Dr.ir. S.N. Paus-Buzink

Department	Applied Ergonomics and Design
-------------------	-------------------------------

R.A. Price

Unit	Industrieel Ontwerpen
Department	Marketing en Consumentenonderzoek
Telephone	+31 15 27 89492

Dr.ir. W.L. Simonse

Unit	Industrieel Ontwerpen
Department	Methodologie en Organisatie van Design
Telephone	+31 15 27 89054
Room	32.B-4-050

Prof.dr. P.J. Stappers

Unit	Industrieel Ontwerpen
Department	Design Conceptualization and Communication
Telephone	+31 15 27 85202
Room	32.C-3-130

Dr. V.T. Visch

Unit	Industrieel Ontwerpen
Department	Design Aesthetics
Telephone	+31 15 27 84951
Room	32.C-3-260